CHANDU PRASAD BHAIRAPU

AI Engineer

bhairapuchanduprasad@gmail.com (+49) 15560194877 linkedin github

Universität Hamburg
Hub of Computing and Data Science (HCDS)
Prof. Dr. Chris Biemann, Dr. Martin Semmann
Albert-Einstein-Ring 8-10, 22761 Hamburg

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Application for Research Associate as Research Software Engineer

Dear Prof. Biemann, dear Dr. Semmann,

I am writing to apply for the Research Associate as Research Software Engineer position at the Hub of Computing and Data Science of Universität Hamburg. I have a Master's in Artificial Intelligence and three years of experience building and deploying AI applications, much of it together with domain experts such as chemists and SAP specialists.

At Kniit, a start-up working for the German energy sector, I am the only developer of an agentic knowledge-extraction pipeline and report directly to the CTO. I built a LangGraph multi-agent system that extracts entities and relationships from complex regulatory and technical documents into a knowledge graph, and evaluated it against a ground truth derived from SAP system data: precision and recall went from 33% and 45% to 93% and 90%. I also built a React and Python web app on Azure where domain experts validate and approve changes to the graph, and an MCP service that makes the graph available for millisecond-latency retrieval.

My master's thesis at Covestro was about evaluating RAG systems. I automated generation and retrieval evaluation with RAGAS and PytrecEval against ground truth written by chemists, ran it on serverless AWS infrastructure built with CDK, and benchmarked multilingual embedding models on more than a million chunks of internal chemical data. Before that, as a data science intern, I compared Random Forest, XGBoost and LSTM models on process time series and gave the chemists a Streamlit dashboard to try parameter changes before running a batch. As an AI consulting intern I then worked with business stakeholders to understand their decision workflows, translated their requirements into a solution approach, and built the agentic LangGraph application: a FastAPI backend, containerized with Docker, deployed to AWS and versioned in GitLab with CI/CD.

HCDS helping a classicist read gravestone inscriptions with AI is the kind of work I want to do: a researcher brings a domain question, and the job is to understand what they need, build the tool, and measure whether it works. Transformer models, RAG and multi-agent setups, and knowledge-based systems are where my experience is strongest; Elasticsearch is the item on your list I have not used yet. I have also published one conference paper, on autonomous indoor vehicles, and the research part of the role is one I would like to grow into.

I will be based in Hamburg, hold an EU Blue Card and am available from 1 January 2027 or earlier. Thank you for considering my application.

Kind regards,

Chandu Prasad Bhairapu